D07 · Warehouse Automation & AMR/Robotics

01 · AMR & mobile robotics: autonomous mobile robots in the warehouse

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Autonomous mobile robots (AMRs) navigate the warehouse with sensors and maps, without fixed guides, to assist picking — bringing totes or shelves to people, following pickers, or moving loads between zones.

2 WHY IT MATTERS

AMRs attack the largest waste in manual picking — walking — without rebuilding the facility. **Because fleets can grow, shrink or move between zones, they suit operations whose volume or layout is still changing.** Their performance depends on WMS integration, good slotting and clear safety rules for working alongside people.

3 HOW IT IS DONE

1. **Define the use case.** Goods-to-person, follow-me or transport, matched to the order profile.
2. **Build the business case with full TCO.** Robots, software, integration, maintenance and the productivity actually achieved in a pilot.
3. **Integrate with the WMS.** Mission assignment driven by real order priorities.
4. **Set safety rules and scale gradually.** Speed limits, zones and procedures; add robots as volume grows.

IN PRACTICE

Collaborative AMRs have the fastest ROI of all automation technologies

Collaborative AMRs don’t require warehouse redesign or the infrastructure of complete GTP systems. They are implemented in weeks, not months.

WMS-AMR integration is critical for optimal robot routing

The WMS must communicate to the Fleet Management Software which picking missions are available so the software assigns the correct AMR to each mission based on its position and mission priority.

AMR and human operator coexistence requires clear safety rules

AMRs must operate in zones where operators know the coexistence rules (maximum speeds, AMR zone signage, emergency protocols).

Scale the AMR fleet with business growth — flexibility is the greatest advantage

AMRs can be added or redistributed between zones in hours — the greatest advantage vs. fixed systems in environments with seasonal demand or rapid growth.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Collaborative AMR implementation — e-commerce DC, 8,200 lines/day — 18 Goods-to-Operator AMRs for picking of the 2,400 highest-velocity SKUs.

Metric	Before (manual picking)	After (AMR-assisted picking)
Picking operator LPPH	OK	64 LPPH -> 168 LPPH (+163%)
Operator walking distance/day	OK	14.2 km/day -> 2.8 km/day (-80%)
Picking errors (incorrect lines)	OK	1.4% error rate -> 0.3% error rate (-79%)

Metric	Before (manual picking)	After (AMR-assisted picking)
Operators required for volume	OK	42 operators -> 18 operators + 18 AMRs

Result: ROI at 24 months: $1.8M in labor savings + $420K in picking error reduction vs. $1.4M investment in 18 AMRs. Payback: 14 months. Picking error reduction was the least expected but most valuable benefit.

HOW IT IS MEASURED

AMR-Assisted LPPH (productivity of operators working with AMRs)

AMR-Assisted LPPHTotal lines picked by operators with AMR / Total hours of AMR-assisted operators

Benchmark: 150–250 LPPH for Goods-to-Operator AMRs · 2–4× manual picking LPPH AMR-Assisted LPPH <120 for a GTO system indicates configuration problems, slotting quality issues, or operator learning curve problems.

AMR Fleet Utilization %

AMR Fleet Utilization %(AMR hours on active mission / Total available hours of the AMR fleet) × 100

Benchmark: >70% fleet utilization in mature operations Low utilization indicates more robots than needed for current volume — or the mission assignment process is not optimized.

WHAT YOU WOULD USE

AMR Providers

Locus Robotics / other collaborative AMR vendors — Module: Collaborative AMR for Order PickingAmong the most widely used collaborative AMR providers in North American e-commerce.

Geek+ / HAI Robotics — Module: Goods-to-Person AMR + ASRSChinese-origin AMR providers with high-density GTP systems.

4 SO WHAT

Use AMRs to remove walking without rebuilding the building. Pick the right use case, prove productivity in a pilot, integrate with the WMS and grow the fleet with demand.

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